

KERALAMMATCH

Privacy-First Matrimony for Kerala & the Global Malayali Community

Master Software Architecture & Product Presentation · Version 2.2

Consent-Based Reveal

24h Time-Bound Access

Multi-Layer Trust

Mobile, Selfie & Govt ID

AI Matching Engine

7-Dimension Scoring %

Clean Architecture

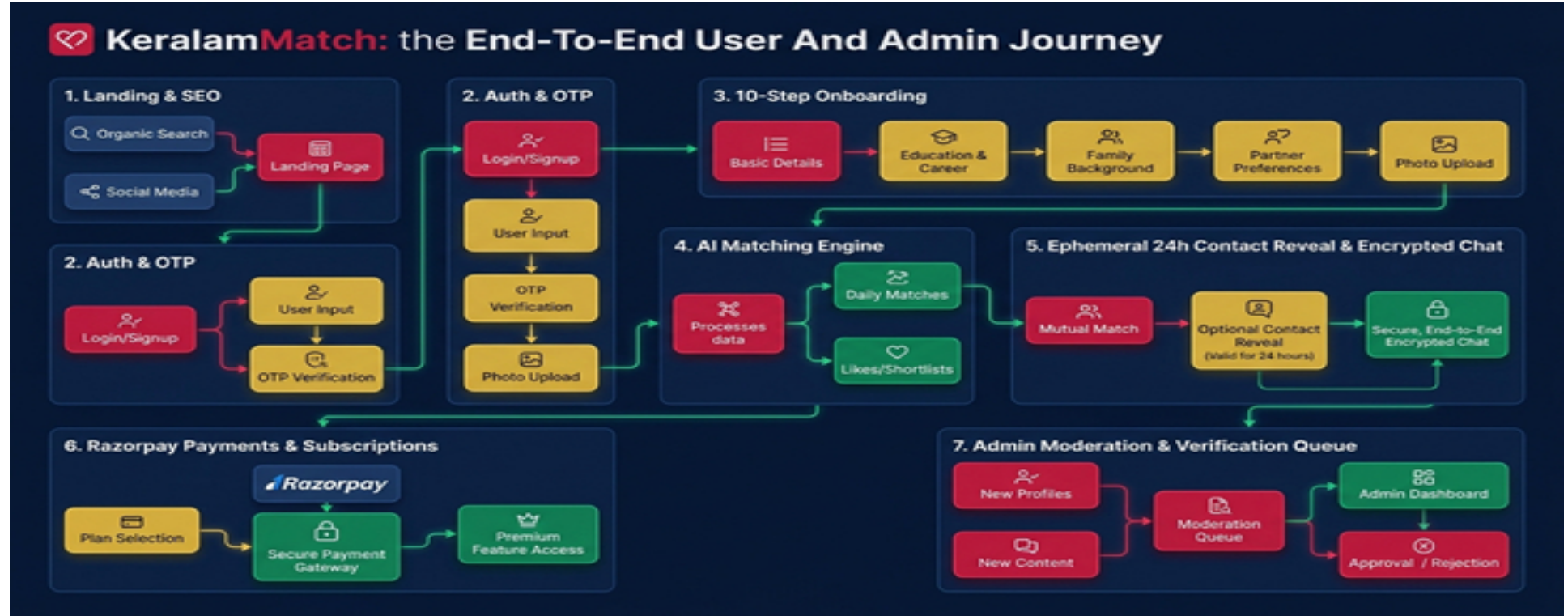
Next.js 16 · Prisma · Razorpay

One Trusted Journey from Discovery to Connection

Discover → Verify → Match → Consent → Communicate → Protect

■ Privacy First Phone & email encrypted with AES-256-GCM at rest; access strictly consent-controlled.	■■ Trust Built-In Multi-layer verification: SMS OTP, email link, Cloudinary selfie liveness & ID queue.	■ AI-Assisted Discovery 7-dimension weighted compatibility scoring (%) and conversational Smart Search.
■■ Controlled Reveal 24-hour time-bound contact window with automatic server-side re-locking.	■ Encrypted Chat Private in-app messaging with read receipts, user blocking & admin privacy shields.	■■■ Operational Governance Admin portal for verification approvals, moderation, revenue ledger & audit logs.

KeralamMatch — End-to-End User & Admin Journey



Customer Journey: New vs. Returning Members

■ New User Journey • Discovery: Arrives via Landing (/), District SEO, or Blog. • Auth: Enters +91 Phone → Verifies SMS OTP → Sets session. • Onboarding: Completes 10-step wizard at /join. • Media (Step 9): Uploads photos to Cloudinary & voice intro. • Verification (Step 10): Submits selfie liveness & ID. • Dashboard: Unlocks recommendations & profile strength. ■ Returning / Registered User • Secure Login: Signs in at /auth → Validates active session. • Status Check: Complete profile → Skips onboarding → Dashboard. • Incomplete Resume: Resumes from last saved step. • Discovery: Filters candidates on /find & AI Smart Search. • Interaction: Shortlists, sends interest, or contact request. • Reveal & Chat: Consented unlock triggers 24h timer & chat.

From Secure Login to a Complete Profile

■ Authentication (/auth) • Firebase Phone Auth (SMS OTP +91) • Google OAuth 2.0 Provider • Encrypted HTTP-only Cookie (`km_session`) • Developer Sandbox OTP guarded in production • Role-based server session validation	■ 10-Step Wizard (/join) 1. Basic Info & DOB 2. Physical Attributes 3. Religion & Caste 4. Education & Career 5. Location Details 6. Family Background 7. Lifestyle Habits 8. Partner Preferences 9. Media Upload ■ (Cloudinary gallery & voice) 10. Verification Submit ■ (Selfie liveness & ID)
--	---

7-Dimension Compatibility Matching + Smart Search

1. Religion & Caste (100) 90 pts same religion; +10 same caste.	2. Education & Career (100) Compares occupational & degree levels.	3. Lifestyle (100) Dietary habits & travel cues.	4. Location (100) District (100), State (75), Other (50).
5. Family Values (100) Marital background alignment.	6. Career Goals (100) GCC / abroad relocation alignment.	7. Communication (100) Base communication compatibility.	■ Smart Search Natural language to structured query.

MULTI-STAGE VERIFICATION SYSTEM

Trust Built Into Every Profile

■ Mobile Verified Firestore SMS Phone OTP	✉ ■ Email Verified Validation Link via Resend	■ Selfie Verified Facial Liveness Capture	■ Identity Verified Aadhaar / ID Review Queue
■ Career Verified Work Email / ID Document	■ Income Verified Tax Statement / Pay Slip	■ Address Verified Utility Bill / Domicile	■ Operational Queue Admin Review ➔ Verified Badge

Privacy Is the Product: 24-Hour Ephemeral Reveal

1. Request Sent Candidate A requests contact. Status: PENDING .	2. Consent Granted Candidate B accepts request. Status: ACCEPTED .	3. Active Reveal Server sets expiresAt = now + 24h. 24:00:00 Countdown.	4. Auto-Locked When t > 24h, API returns 403. Status: EXPIRED .
---	--	---	---

■ **Cryptographic & Security Enforcement**

- Phone & email are encrypted at rest with AES-256-GCM in PostgreSQL.
- The 24-hour timer applies strictly to contact credential reveal — NOT to the candidate profile.
- Server-Side Authority: Browser countdown is a visual aid; server API verifies expiry on every access.
- Secure In-App Chat is an independent capability and remains active after reveal expiration.

Secure In-App Messaging & Privacy Shields

■ Messaging Capabilities

- Split-pane messaging layout (/chat)
- AES-256-GCM message encryption at rest
- Read receipts & double checkmarks (✓✓)
- Instant one-click user blocking & reporting
- Rate limiting: 30 msgs/min via Upstash Redis

■■ Admin Chat Privacy Shield

- **No Global Chat Reading:** Admins cannot view private chats.
- **Repository Constraints:** Requires session.user.id as participant.
- **Reported Message Redaction:** Inspection restricted to official abuse reports.
- **Zero Side-Channel Leakage:** Audit logs exclude private chat text.

Premium Membership & Secure Payments

Free Tier ■0 (Forever) Standard browsing, receive contact requests.	Silver Plan ■999 / mo Unlimited chat, 5 verified reveals, basic astro.	Gold Plan (Popular) ■1,999 / 3 mos 15 reveals, priority ranking, priority support.	Platinum Plan ■3,499 / yr 30 reveals, relationship advisor, top boost.
--	---	---	---

■ **Razorpay Payment & Webhook Verification Flow**

Plan Selection ➔ Server Action creates Razorpay Order ➔ Client Modal Checkout (UPI/Cards) ➔ Webhook (‘payment.captured’) ➔ Server validates HMAC SHA-256 Signature ➔ PostgreSQL updates Subscription / Wallet balance ➔ Receipt & tax invoice sent via Resend.

Pay-Per-Reveal Wallet: ■99 (1 Reveal) and ■499 (5 Reveals) allow flexible on-demand unlocks without subscriptions.

The KeralamMatch Customer Platform

■ Public & SEO • Landing Page (/) • 13 District SEO Hubs • 6 Community SEO Hubs • Blog Magazine (/blog)	■ Auth & Onboard • Mobile SMS OTP Auth • Google Sign-In • 10-Step Wizard (/join) • Cloudinary Photo Gallery	■ Member Portal • Dashboard (/dashboard) • Find Matches (/find) • Profile View (/profile/[id]) • Shortlists & Bookmarks	■ Chat & Commerce • Encrypted Chat (/chat) • Contact Requests (/requests) • Notifications (/notifications) • Pricing & Plans (/pricing)
--	--	--	--

Powerful Governance Without Compromising Privacy

1. Admin Login (/admin/login) Guarded entry gate with session checks.	2. Master Dashboard (/admin) Executive KPIs & registration charts.	3. User Management (/admin/users) Directory search & ban/verify controls.	4. Verification Queue (/admin/verification) Selfie liveness & ID document review.	5. Reports (/admin/reports) Safety moderation & complaint reviews.
6. Payments (/admin/payments) Revenue ledger, stats & refunds.	7. Blog CMS (/admin/blog) Create, edit & publish SEO guides.	8. FAQ Management (/admin/faq) Customer support Q&A; editor.	9. Audit Logs (/admin/audit) Immutable timeline of staff actions.	10. Settings (/admin/settings) Platform rules & maintenance mode.

Built for Secure, Scalable Web Delivery

■■ Layered Clean Architecture

- **Presentation:** Next.js 16 App Router (RSC + Client Components) + Tailwind CSS v4
- **Controller Layer:** Server Actions & Route Handlers with Zod schema validation
- **Service Layer:** 10 domain modules (Auth, Profile, Matching, Messaging, Contact, etc.)
- **Repository Layer:** Isolated domain database queries
- **Persistence:** PostgreSQL on Neon Serverless + Prisma ORM v6

■■ Managed Cloud Integrations

- **Firebase Auth:** SMS Phone OTP (+91) & Google OAuth 2.0
- **Cloudinary CDN:** AVIF/WebP image optimization, watermarking, audio intros
- **Razorpay Gateway:** Subscriptions, wallet recharges, HMAC webhooks
- **Upstash Redis:** Sliding-window rate limiting & queues
- **Resend Email:** Branded transactional HTML emails
- **Vercel Edge:** Global serverless hosting (Singapore sin1)

Security, Quality & Mobile Readiness

■ Security Hardening • AES-256-GCM field encryption for phone & email • HTTP-only cookie `km_session` prevents XSS • Upstash Redis rate limiting on auth, chat & admin • Content Security Policy, HSTS & X-Frame-Options • HMAC SHA-256 Razorpay webhook validation	■ QA Test Baseline • 24 / 24 Unit Tests Passed (100% Pass Rate) • 8 / 8 E2E Flow Scenarios Passed (100% Pass) • 53 / 53 Routes Compiled Cleanly • 0 TypeScript compiler errors (strict mode) • 0 ESLint audit errors	■ Mobile & PWA Strategy • Responsive verified: 360px, 390px, 430px, 768px, 1440px • Web App Manifest (manifest.json) configured • Service Worker (sw.js) for asset caching & offline • Single responsive web codebase ready for PWA • Future Android TWA / Play Store packaging path
--	--	---

Ready for Controlled Soft Launch

Day 1: Prod DB Setup Neon DB & migrations	Day 2: Customer QA Register, 10-step & search	Day 3: 24h Reveal QA Consent, countdown & chat	Day 4: Payments QA Razorpay & webhooks	Day 5: Admin Live Data Prisma aggregate queries	Day 6: PWA & SEO Sitemap & mobile checks	Day 7: Soft Launch Domain switch & launch
---	---	--	--	---	--	---

KeralamMatch

Discover → Verify → Match → Consent → Communicate → Protect

Privacy First. Verified Identity. Cultural Relevance. Controlled 24-Hour Ephemeral Reveal.